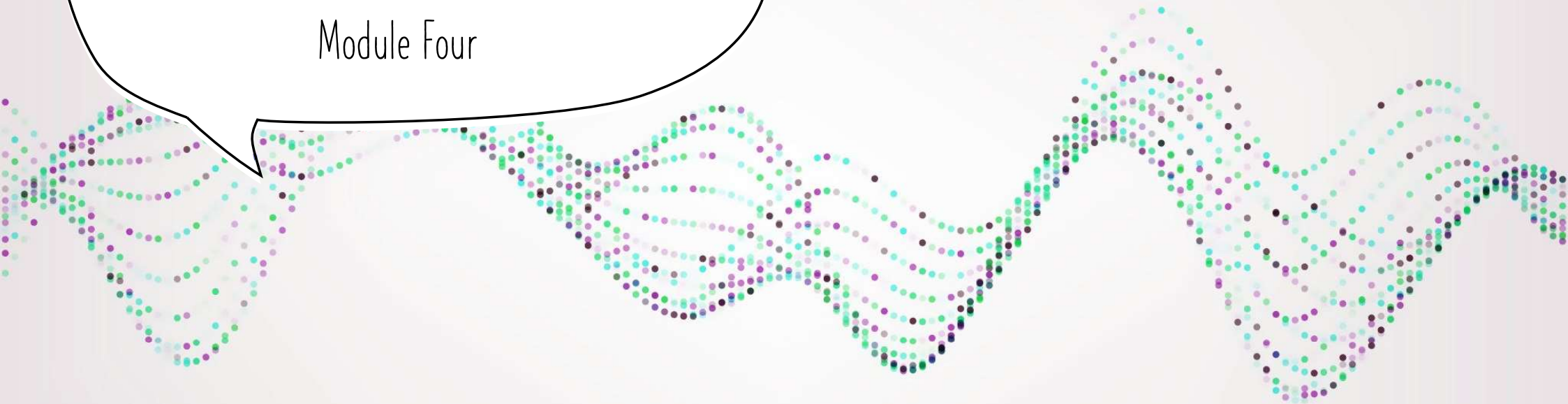


NETWORK FLOW

Module Four



OUTLINE

Single Source shortest path: Bellman Ford Algorithm,

All pair shortest path: Floyd-Warshall Algorithm

Network Flows: Flow Networks,

Maximum Flows:

- Ford-Fulkerson,
- Edmond-Karp,
- Push Re-label Algorithm

Application of Max Flow to Maximum Matching Problem

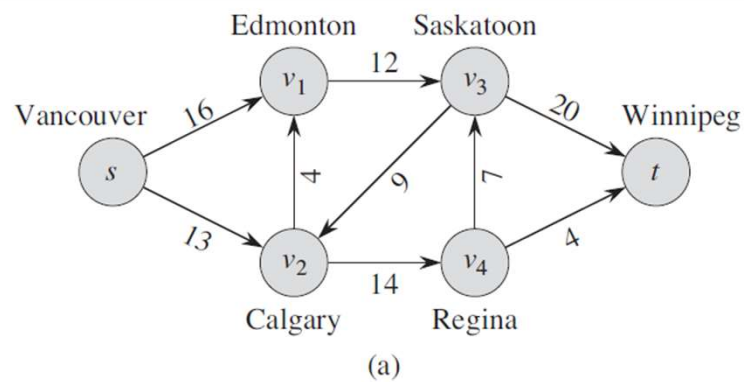
WHAT IS NETWORK FLOW ?

Flow network is a directed graph $G=(V,E)$ such that each edge has a non-negative capacity $c(u,v) \geq 0$.

Two distinguished vertices exist in G namely :

- Source (denoted by s) : In-degree of this vertex is 0.
- Sink (denoted by t) : Out-degree of this vertex is 0.

Flow in a network is an integer-valued function f defined on the edges of G satisfying $0 \leq f(u,v) \leq c(u,v)$, for every edge (u,v) in E .



WHAT IS NETWORK FLOW ?

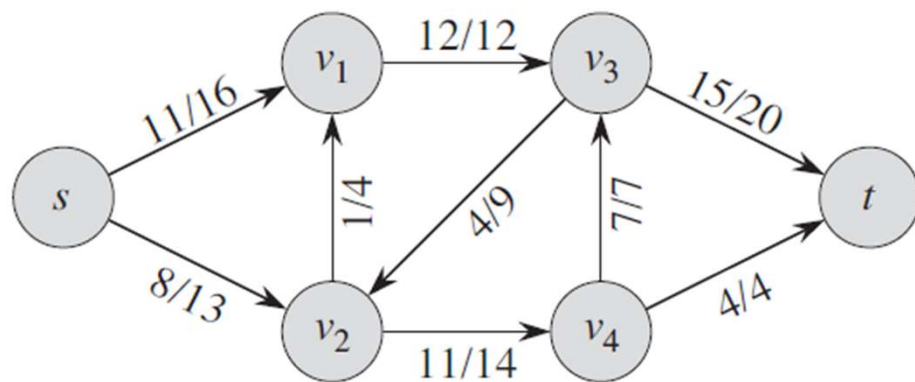
- Each edge (u,v) has a non-negative capacity $c(u,v)$.
 - If (u,v) is not in E assume $c(u,v)=0$.
 - We have source s and sink t .
- Assume that every vertex v in V is on some path from s to t .

CONDITIONS FOR NETWORK FLOW

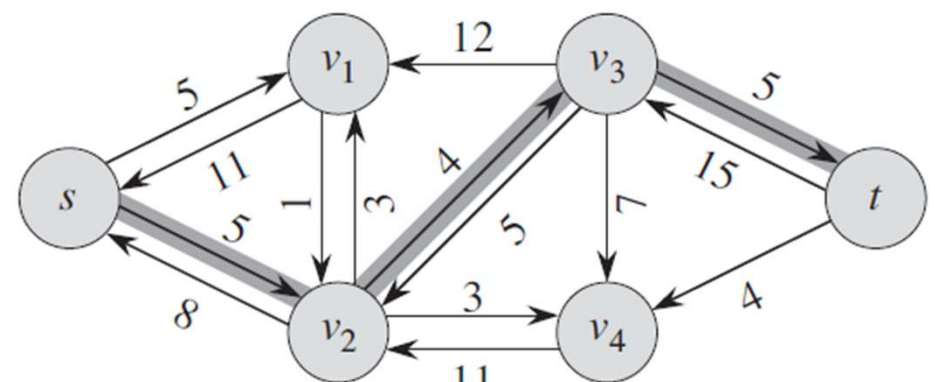
For each edge (u,v) in E , the flow $f(u,v)$ is a real-valued function that must satisfy the following 3 conditions :

- Capacity Constraint : $\forall u,v \in V, f(u,v) \leq c(u,v)$
- Skew Symmetry : $\forall u,v \in V, f(u,v) = -f(v,u)$
- Flow Conservation: $\forall u \in V - \{s,t\} \sum_{v \in V} f(u,v) = 0$

Skew symmetry condition implies that $f(u,u)=0$.



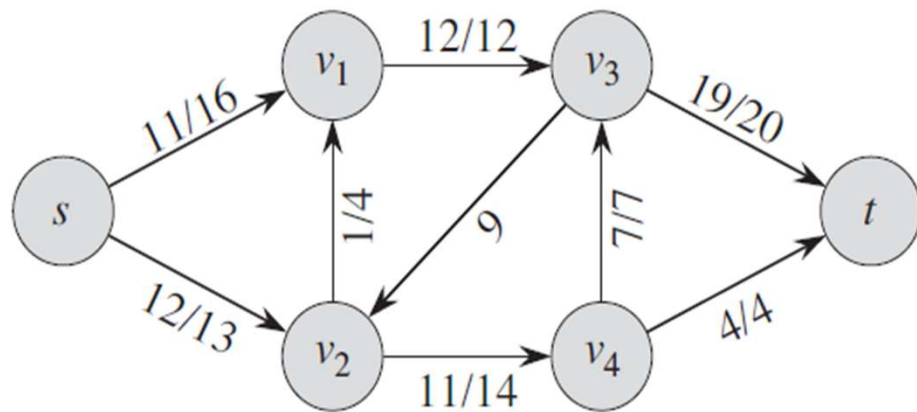
(a)



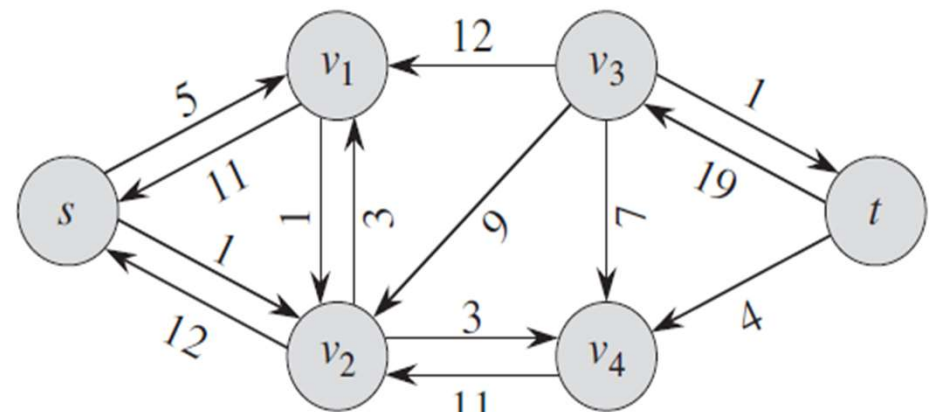
(b)

EXAMPLE

AUGMENTING ALONG PATH P BY ITS RESIDUAL CAPACITY 4.



(c)



(d)

If f is a flow in a flow network $G = (V, E)$ with source s and sink t , then the following conditions are equivalent:

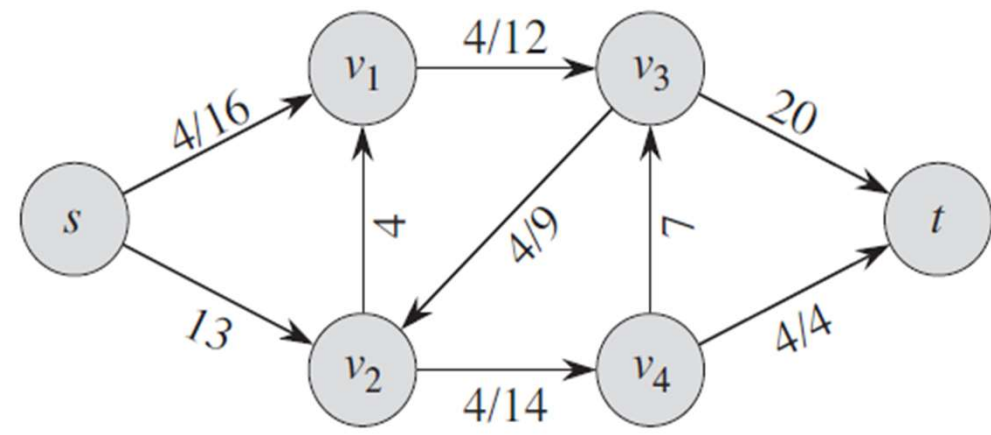
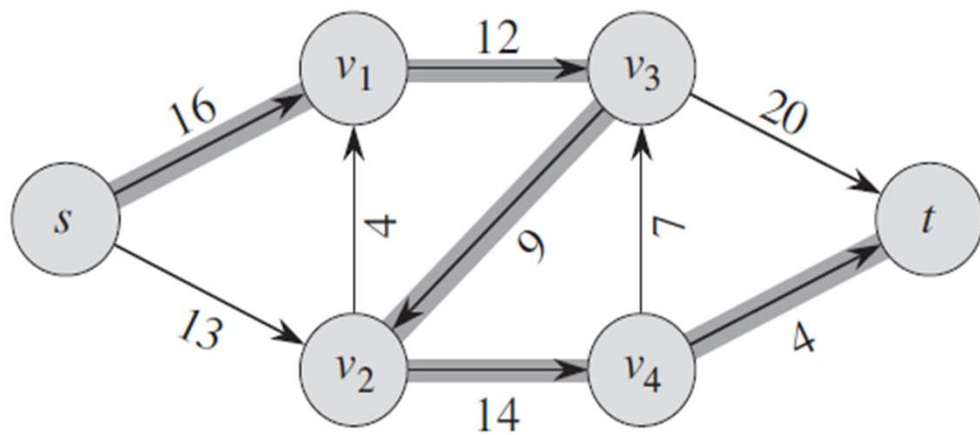
1. f is a maximum flow in G .
2. The residual network G_f contains no augmenting paths.
3. $|f| = c(S, T)$ for some cut (S, T) of G .

MAX FLOW MIN CUT THEOREM

FORD-FULKERSON(G, s, t)

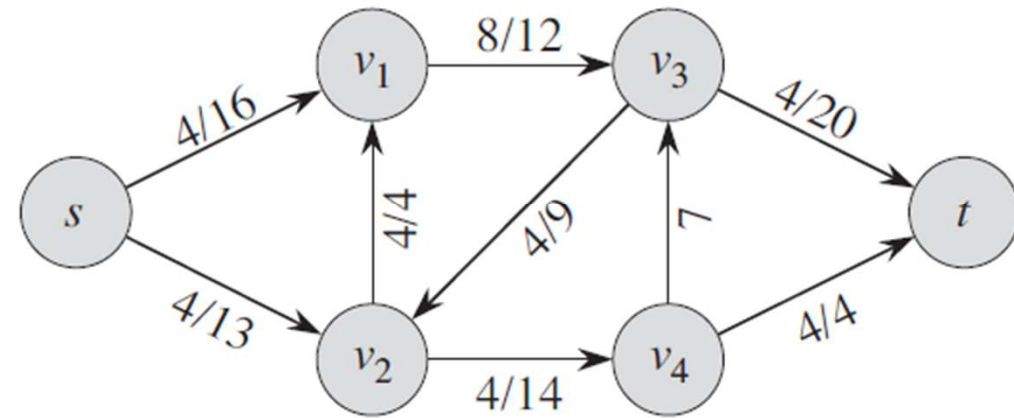
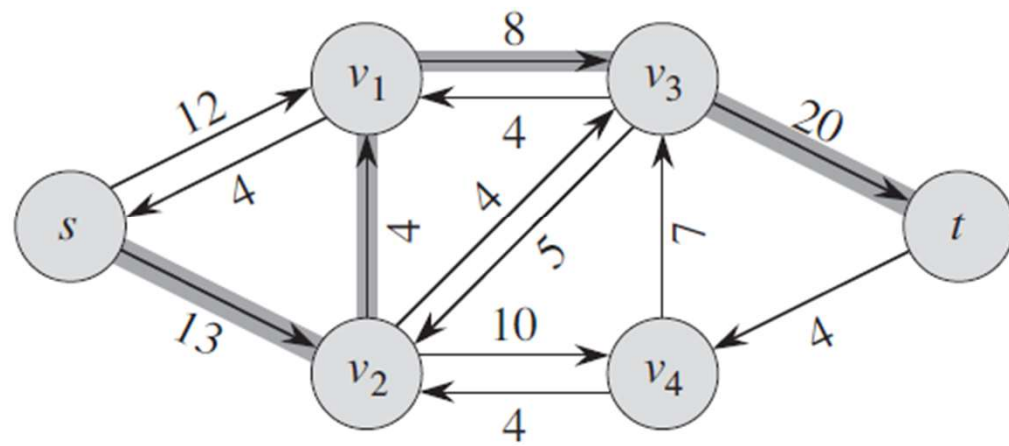
```
1  for each edge  $(u, v) \in G.E$ 
2       $(u, v).f = 0$ 
3  while there exists a path  $p$  from  $s$  to  $t$  in the residual network  $G_f$ 
4       $c_f(p) = \min \{c_f(u, v) : (u, v) \text{ is in } p\}$ 
5      for each edge  $(u, v)$  in  $p$ 
6          if  $(u, v) \in E$ 
7               $(u, v).f = (u, v).f + c_f(p)$ 
8          else  $(v, u).f = (v, u).f - c_f(p)$ 
```

FORD-FULKERSON METHOD

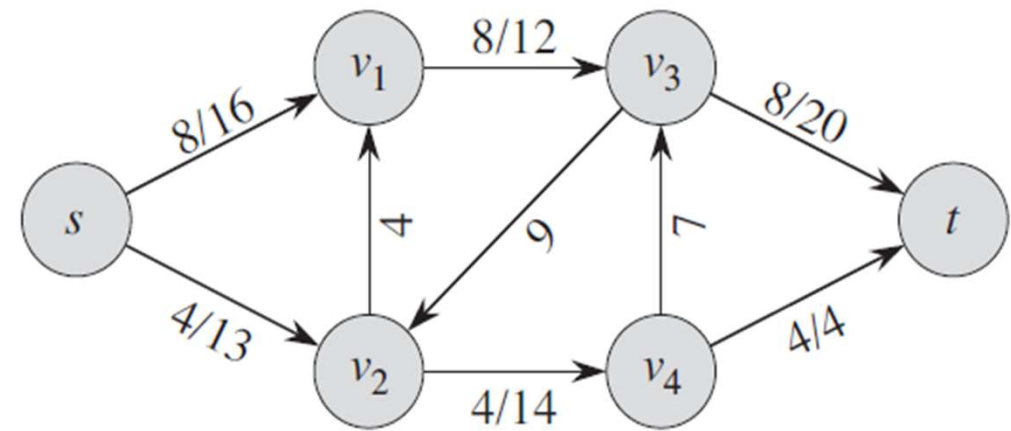
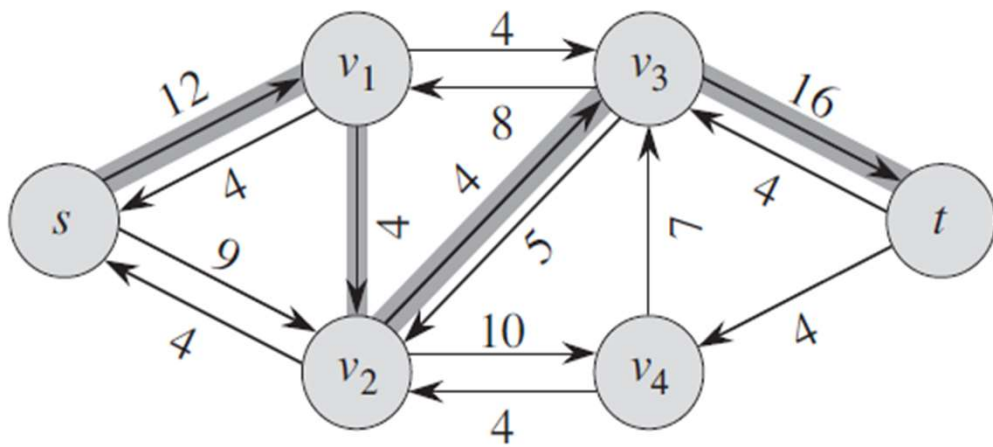


FORD-FULKERSON ALGORITHM

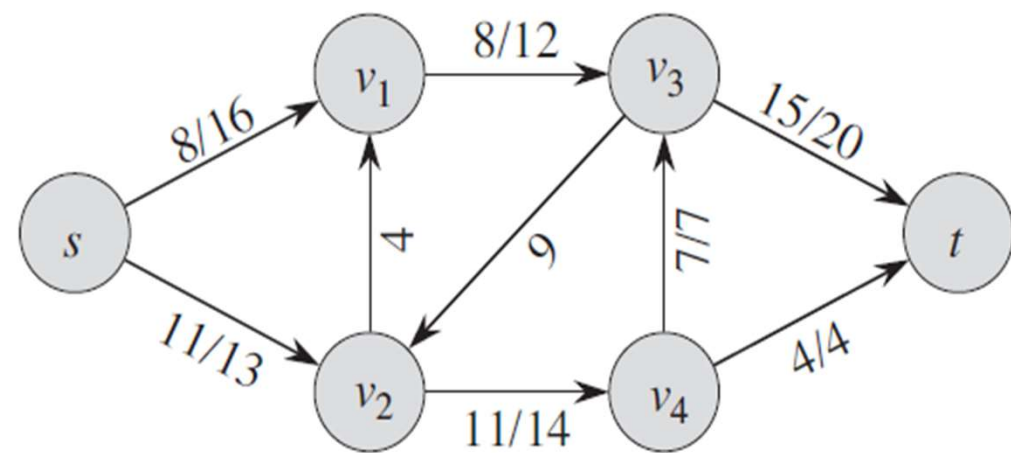
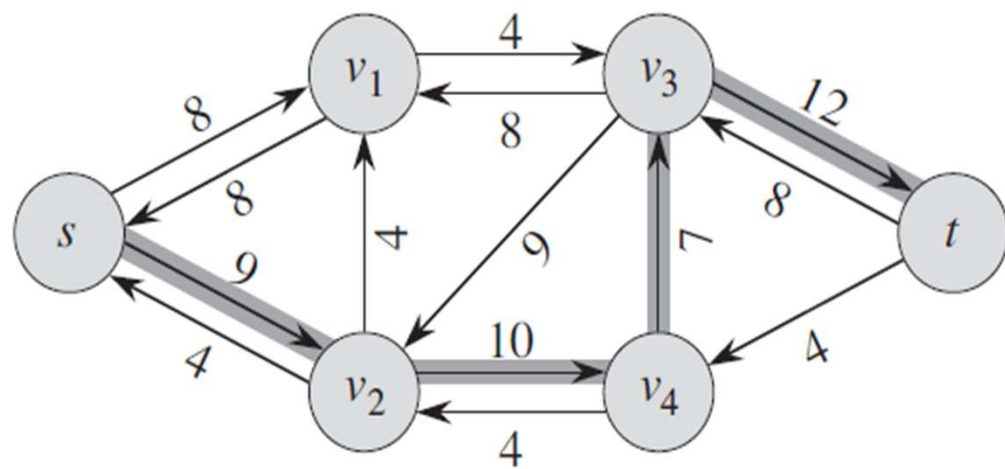
PATH $s \rightarrow v_1 \rightarrow v_3 \rightarrow v_2 \rightarrow v_4 \rightarrow t$



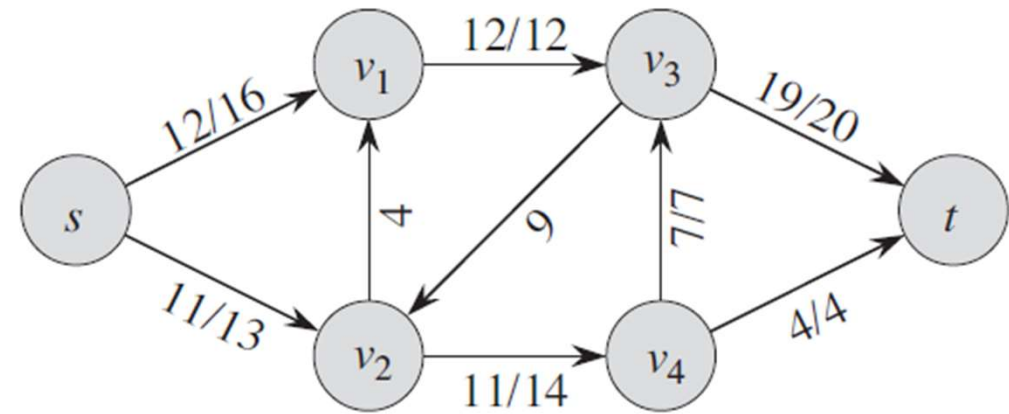
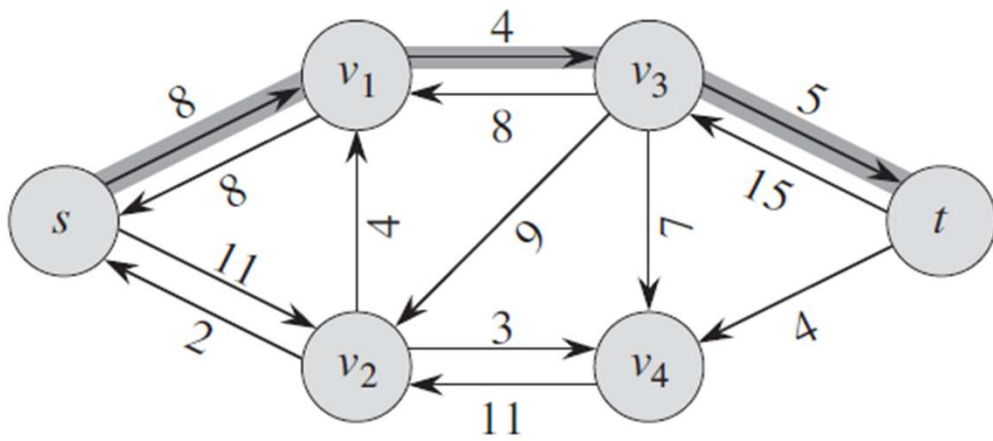
PATH $S \rightarrow V_2 \rightarrow V_1 \rightarrow V_3 \rightarrow T$



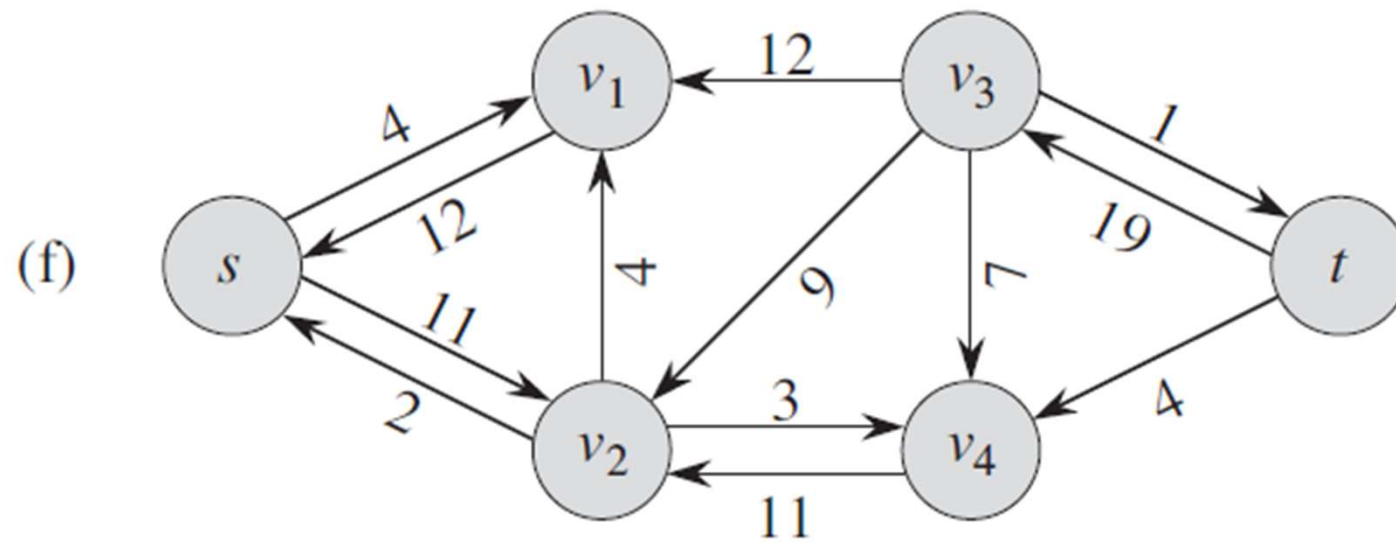
PATH: $s \rightarrow v_1 \rightarrow v_2 \rightarrow v_3 \rightarrow t$



PATH: $s \rightarrow v_2 \rightarrow v_4 \rightarrow v_3 \rightarrow t$



PATH: $s \rightarrow v_1 \rightarrow v_3 \rightarrow t$



RESIDUAL NETWORK